

Worksheet No. – 1

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Aim/Overview of the practical:

Create a worksheet in MS Excel having student records. Demonstrate the use of COUNT, COUNTA, COUNTBLANK, SUM, AVERAGE, IF, NESTEDIF, SUMIF, AVERAGEIF, IFERROR, COUNTIF, AND and OR.

Input/Apparatus Used:

- Computer / Laptop
- MS Excel

Procedure:

- Open **Microsoft Excel**.
- Create a new workbook.
- Enter sample data such as Student Name, Marks, Attendance, Department, etc.
- Select an empty cell where you want the result.
- Type the required formula.
- Press **Enter** to display the result.

Formula I have Used:

- COUNT:- Counts only cells containing numbers.

SYNTAX:- =COUNT(value1, value2,...)

A12			
=COUNT(A2:A11)			
	A	B	C
1	UID	Name	Marks
2	101	Saloni Singh	98
3	102	Kriti Dubey	96
4	103	Anjali Singh	90
5	104	Manshi Singh	98
6	105	Muskan Singh	97
7	106	Prachi Raj	99
8	107	Shivam Singh	94
9	108	Satyam Singh	98
10	109	Shubham Singh	98
11	110	Thakur Rajveer Singh	90
12	10		

- COUNTA:- Counts all non-empty cells (numbers, text, logical values).

SYNTAX:- =COUNTA(value1, value2,...)

C12 X ✓ fx =COUNTA(C2:C11)			
	A	B	C
1	UID	Name	Marks
2	101	Saloni Singh	98
3	102	Kriti Dubey	96
4	103	Anjali Singh	90
5	104	Manshi Singh	98
6	105	Muskan Singh	97
7	106	Prachi Raj	99
8	107	Shivam Singh	A
9	108	Satyam Singh	98
10	109	Shubham Singh	98
11	110	Thakur Rajveer Singh	90
12	10		10

- COUNTBLANK:- Counts the number of empty cells in a range.

SYNTAX:- =COUNTBLANK(range)

C12 X ✓ fx =COUNTBLANK(C2:C11)				
	A	B	C	Depart
1	UID	Name	Marks	
2	101	Saloni Singh	98	MCA
3	102	Kriti Dubey	96	MCA
4	103	Anjali Singh	90	MCA
5	104	Manshi Singh	98	MCA
6	105	Muskan Singh	97	MCA
7	106	Prachi Raj	99	MCA
8	107	Shivam Singh		BCA
9	108	Satyam Singh	98	MCA
10	109	Shubham Singh	98	MCA
11	110	Thakur Rajveer Singh	90	MCA
12	10			1

- SUM:- Adds all numbers in a range.

SYNTAX:- =SUM(number1, number2,...)

C12 X ✓ fx =SUM(C2:C11)			
	A	B	C
1	UID	Name	Marks
2	101	Saloni Singh	98
3	102	Kriti Dubey	96
4	103	Anjali Singh	90
5	104	Manshi Singh	98
6	105	Muskan Singh	97
7	106	Prachi Raj	99
8	107	Shivam Singh	95
9	108	Satyam Singh	98
10	109	Shubham Singh	98
11	110	Thakur Rajveer Singh	90
12	10		959

- AVERAGE:- Returns the average (mean) of numbers.

SYNTAX:- =AVERAGE(number1, number2,...)

C12 X ✓ fx =AVERAGE(C2:C11)			
	A	B	C
1	UID	Name	Marks
2	101	Saloni Singh	98
3	102	Kriti Dubey	96
4	103	Anjali Singh	90
5	104	Manshi Singh	98
6	105	Muskan Singh	97
7	106	Prachi Raj	99
8	107	Shivam Singh	95
9	108	Satyam Singh	98
10	109	Shubham Singh	98
11	110	Thakur Rajveer Singh	90
12	10		95.9

- **IF:-** Performs a logical test and returns one value if TRUE and another if FALSE.

SYNTAX:- =IF(logical_test, value_if_true, value_if_false)

G2 :  =IF(F2>80,"Yes","No")							
	A	B	C	D	E	F	G
1	UID	Name	Marks	Department	Fee Status	Attendance	Eligible
2	101	Saloni Singh	98	MCA	Paid	90	Yes
3	102	Kriti Dubey	96	MCA	Paid	87	Yes
4	103	Anjali Singh	90	MCA	Unpaid	78	No
5	104	Manshi Singh	98	MCA	Paid	87	Yes
6	105	Muskan Singh	97	MCA	Unpaid	76	No
7	106	Prachi Raj	99	MCA	Paid	89	Yes
8	107	Shivam Singh	95	BCA	Unpaid	86	Yes
9	108	Satyam Singh	98	MCA	Paid	78	No
10	109	Shubham Singh	98	MCA	Unpaid		No
11	110	Thakur Rajveer Singh	90	MCA	Paid	80	No

- **Nested IF:-** Uses multiple IF statements to test multiple conditions.

SYNTAX:- =IF(condition1,result1,IF(condition2,result2,result3))

D2 :  =IF(C2>96,"A+",IF(C2>=95,"A","B+"))					
	A	B	C	D	E
1	UID	Name	Marks		Department
2	101	Saloni Singh	98	A+	MCA
3	102	Kriti Dubey	96	A	MCA
4	103	Anjali Singh	90	B+	MCA
5	104	Manshi Singh	98	A+	MCA
6	105	Muskan Singh	97	A+	MCA
7	106	Prachi Raj	99	A+	MCA
8	107	Shivam Singh	95	A	BCA
9	108	Satyam Singh	98	A+	MCA
10	109	Shubham Singh	98	A+	MCA
11	110	Thakur Rajveer Singh	90	B+	MCA

- **SUMIF:-** Adds cells that meet a specified condition.

SYNTAX:- =SUMIF(range, criteria, [sum_range])

C12 :  =SUMIF(F2:F11,"paid",C2:C11)						
	A	B	C	D	E	F
1	UID	Name	Marks		Department	Fee Status
2	101	Saloni Singh	98	A+	MCA	Paid
3	102	Kriti Dubey	96	A	MCA	Paid
4	103	Anjali Singh	90	B+	MCA	Unpaid
5	104	Manshi Singh	98	A+	MCA	Paid
6	105	Muskan Singh	97	A+	MCA	Unpaid
7	106	Prachi Raj	99	A+	MCA	Paid
8	107	Shivam Singh	95	A	BCA	Unpaid
9	108	Satyam Singh	98	A+	MCA	Paid
10	109	Shubham Singh	98	A+	MCA	Unpaid
11	110	Thakur Rajveer Singh	90	B+	MCA	Paid
12	10		579			

- **AVERAGEIF**:- Finds the average of cells that meet a condition.

SYNTAX:- =AVERAGEIF(range, criteria, [average_range])

C12 :    =AVERAGEIF(F2:F11,"paid",C2:C11)						
	A	B	C	D	E	F
1	UID	Name	Marks		Department	Fee Status
2	101	Saloni Singh	98 A+		MCA	Paid
3	102	Kriti Dubey	96 A		MCA	Paid
4	103	Anjali Singh	90 B+		MCA	Unpaid
5	104	Manshi Singh	98 A+		MCA	Paid
6	105	Muskan Singh	97 A+		MCA	Unpaid
7	106	Prachi Raj	99 A+		MCA	Paid
8	107	Shivam Singh	95 A		BCA	Unpaid
9	108	Satyam Singh	98 A+		MCA	Paid
10	109	Shubham Singh	98 A+		MCA	Unpaid
11	110	Thakur Rajveer Singh	90 B+		MCA	Paid
12	10		96.5			

- **IFERROR**:- Returns a custom value if a formula results in an error.

SYNTAX:- =IFERROR(value, value_if_error)

C12 :    =IFERROR(C2/G2,"Error")							
	A	B	C	D	E	F	G
1	UID	Name	Marks		Department	Fee Status	Attendance
2	101	Saloni Singh	98 A+		MCA	Paid	90
3	102	Kriti Dubey	96 A		MCA	Paid	87
4	103	Anjali Singh	90 B+		MCA	Unpaid	78
5	104	Manshi Singh	98 A+		MCA	Paid	87
6	105	Muskan Singh	97 A+		MCA	Unpaid	76
7	106	Prachi Raj	95 A		MCA	Paid	89
8	107	Shivam Singh	95 A		BCA	Unpaid	86
9	108	Satyam Singh	98 A+		MCA	Paid	78
10	109	Shubham Singh	98 A+		MCA	Unpaid	
11	110	Thakur Rajveer Singh	90 B+		MCA	Paid	80
12	10		1.0888889				FALSE

- **COUNTIF**:- Counts cells that satisfy a condition.

SYNTAX:- =COUNTIF(range, criteria)

   =COUNTIF(F2:F11,"paid")				
B	C	D	E	F
Name	Marks		Department	Fee Status
Saloni Singh	98 A+		MCA	Paid
Kriti Dubey	96 A		MCA	Paid
Anjali Singh	90 B+		MCA	Unpaid
Manshi Singh	98 A+		MCA	Paid
Muskan Singh	97 A+		MCA	Unpaid
Prachi Raj	95 A		MCA	Paid
Shivam Singh	95 A		BCA	Unpaid
Satyam Singh	98 A+		MCA	Paid
Shubham Singh	98 A+		MCA	Unpaid
Thakur Rajveer Singh	90 B+		MCA	Paid
	1.0888889			6

- **AND:-** Returns TRUE only if all conditions are TRUE.

SYNTAX:- =AND(logical1, logical2,...)

=IF(AND(C2>=90,G2>=80),"Yes","no")						
B	C	D	E	F	G	H
Name	Marks		Department	Fee Status	Attendance	Eligible
Saloni Singh	98	A+	MCA	Paid	90	Yes
Kriti Dubey	96	A	MCA	Paid	87	Yes
Anjali Singh	90	B+	MCA	Unpaid	78	no
Manshi Singh	98	A+	MCA	Paid	87	Yes
Muskan Singh	97	A+	MCA	Unpaid	76	no
Prachi Raj	95	A	MCA	Paid	89	Yes
Shivam Singh	95	A	BCA	Unpaid	86	Yes
Satyam Singh	98	A+	MCA	Paid	78	no
Shubham Singh	98	A+	MCA	Unpaid		no
Thakur Rajveer Singh	90	B+	MCA	Paid	80	Yes

- **OR:-** Returns TRUE if at least one condition is TRUE.

SYNTAX:- =OR(logical1, logical2,...)

=IF(OR(C2>=90,G2>=80),"Yes","no")						
B	C	D	E	F	G	H
Name	Marks		Department	Fee Status	Attendance	Eligible
Saloni Singh	98	A+	MCA	Paid	90	Yes
Kriti Dubey	96	A	MCA	Paid	87	Yes
Anjali Singh	90	B+	MCA	Unpaid	78	Yes
Manshi Singh	98	A+	MCA	Paid	87	Yes
Muskan Singh	97	A+	MCA	Unpaid	76	Yes
Prachi Raj	95	A	MCA	Paid	89	Yes
Shivam Singh	95	A	BCA	Unpaid	86	Yes
Satyam Singh	98	A+	MCA	Paid	78	Yes
Shubham Singh	98	A+	MCA	Unpaid		Yes
Thakur Rajveer Singh	90	B+	MCA	Paid	80	Yes

Learning outcomes:

- Learned to use the IF function to make decisions based on conditions.
- Applied SUM and AVERAGE to calculate the sum & average of values.
- Applied SUMIF to calculate the sum of values that meet a specified condition.
- Applied AVERAGEIF to calculate the average of values that meet a specified condition.
- Used COUNTIF to count cells that satisfy a given criterion.
- Implemented IFERROR to handle errors and display user-friendly messages.
- Used AND to test whether multiple conditions are true simultaneously.
- Used OR to check if at least one of multiple conditions is true.